

Product Requirements Document (PRD)

Reliability-Aware Multi-Agent System for Chest X-Ray Pneumonia Classification

Doc status: Draft v1 **Owner:** [your name] **Last updated:** September 2026

1. Overview

1.1 Problem statement

Standard deep-learning CXR classifiers output a probability for *any* input — a clean, diagnostically clear image and a blurry, out-of-distribution, or genuinely ambiguous one get treated identically. A pneumonia classifier saying "Pneumonia: 61%" doesn't tell a clinician whether that 61% reflects real diagnostic ambiguity or the model simply being confused by bad data. That distinction changes what should happen next — one case needs a second opinion, the other just needs a better X-ray.

1.2 Product goal

Build a pipeline that wraps a pneumonia classifier with reliability-checking logic so the system **knows when to trust its own output** and takes a different, appropriate action (accept / repair / escalate / reject)

depending on how reliable a given input–prediction pair actually is — instead of always emitting a bare number.

1.3 Non-goals

- Not a clinically certified or deployable diagnostic product — this is a research prototype.
- Not attempting to beat state-of-the-art raw classification accuracy; the target metric is system-level reliability, not top-line AUC.
- Not putting a general-purpose LLM in the diagnostic decision path (see §8, literature justification).

1.4 Success metrics

- Accuracy/error rate **broken down by action taken** (Accept / Repair / Escalate / Reject) is measurably better than a single-model baseline on the "Accept" bucket, without silently dumping hard cases.
 - OOD detection AUROC $\geq$ target on a held-out natural OOD set (NIH vs. CheXpert/PadChest).
 - Calibration error (ECE) improves over the raw base-model baseline.
 - Compute cost per input stays within a defined budget for the fast path, with escalation used only when needed.
 - Repair recovery rate (empirically measured, §7) reported honestly, including failure cases on heavy degradation.
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2. Users & Use Cases

User	Need
Researcher / mentor evaluating this project	An explainable, auditable pipeline with clear parameters and a defensible evaluation methodology, not a black box.

Downstream clinical workflow (simulated, not real deployment)	A prediction that's either trustworthy-and-actionable or clearly flagged for human review — never a silently wrong confident answer.
Future engineer extending the system	Modular agents with well-defined I/O contracts that can be swapped, recalibrated, or upgraded (e.g., rule table → learned Decision Agent) independently.

3. System Architecture (Functional Requirements)

Seven agents, each a discrete function: image (or model output) in → **score + reasoning label** out. Only the Base Model is a trained neural network; the rest are lightweight, mostly deterministic components — the "agent" framing describes coordination and decision-making, not that everything is a neural net.

FR-1 — Quality Agent

Detects technical defects before they contaminate the diagnosis.

Defect	Parameter	Method	Flag threshold*
Blur	Laplacian variance	Grayscale → Laplacian filter → variance	< ~100
Noise	SNR (dB)	$20 \cdot \log_{10}(\text{mean_signal} / \text{std_dev}(\text{noise_residual}))$	< ~15 dB
Exposure	Mean intensity + histogram spread	Mean/stddev of 0–255 histogram	mean < 20 or > 235

*All calibrated per dataset via ROC on a held-out split, never assumed.

Derived blur severity score: `blur_% = 100 × (1 - min(laplacian_variance / reference_max_variance, 1))`.

FR-2 — OOD Agent

Detects distribution shift (different scanner, wrong body part, unfamiliar population).

- Primary: Mahalanobis distance of the base model's mid-layer feature vector from the training feature distribution (mean/covariance), thresholded at e.g. the 95th percentile of in-distribution distances.
- Backup/cross-check: Energy score from output logits.
- Natural validation set: NIH (in-distribution for the small model) vs. CheXpert/PadChest (real, not synthetic, OOD) — see §6.

FR-3 — Base Model

The actual pneumonia classifier. See §5 (Tech Stack) for model selection.

FR-4 — Uncertainty Agent

Measures genuine model confidence, not just the predicted label — softmax confidence + predictive entropy, computed purely as post-processing on the base model's own output logits (no separate model).

FR-5 — Decision Agent (orchestrator — the core novel contribution)

Combines quality, OOD, and confidence signals into one action:

Quality	OOD	Confidence	Action
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Good	In-distribution	High	Accept
Poor	In-distribution	—	Repair → re-check
Good	In-distribution	Low	Escalate to larger model
—	Severely OOD	—	Reject → human review

- v1: fixed rule/threshold table (explainable baseline).
- v2 (stretch): small learned weighting function (e.g., logistic regression/shallow MLP) over the three scalar signals, trained to optimize the accuracy/cost/safety trade-off — this is the part to defend as genuinely novel; everything else uses established techniques.

FR-6 — Repair Agent

Applies targeted, reversible fixes only:

Defect	Fix
Blur	Wiener deconvolution / unsharp masking
Noise	Non-local means denoising
Exposure	CLAHE

Hard constraint: never invents missing anatomy; every repaired output is tagged and never trusted at the same level as a clean original.

FR-7 — Verification Agent

Confirms a repair/escalation actually helped, using confidence delta and quality-score delta before vs. after. A fix is accepted only if confidence improves by a meaningful margin (e.g., ≥ 0.15); otherwise the system escalates further rather than trusting a marginal, possibly-noisy improvement.

FR-8 — End-to-end pipeline flow

1. Image arrives → Quality + OOD Agents run in parallel.
 2. Base Model predicts → Uncertainty Agent scores it.
 3. Decision Agent picks Accept / Repair / Escalate / Reject.
 4. If Repair/Escalate fired → Verification Agent re-checks before release.
 5. Output: prediction **with a reliability label**, or a "needs human review" flag.
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4. Non-Functional Requirements

- **Explainability:** every agent must return a human-readable reasoning label alongside its score — no opaque outputs.
 - **Auditability:** every decision (action taken, signals that drove it, before/after verification deltas) must be logged per-inference for later analysis.
 - **Latency budget:** fast path (Accept/Reject without escalation) should run in near real time on CPU or single-GPU; escalation path may be slower.
 - **Reproducibility:** all thresholds, model checkpoints, and dataset splits version-pinned and documented, since thresholds are empirically calibrated, not universal constants.
 - **Safety default:** when in doubt (ambiguous signals near a threshold boundary), the system should fail toward Escalate/Reject, never toward silent Accept.
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5. Tech Stack

Layer	Choice	Notes
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Language	Python 3.10+	Standard for the ML/CV ecosystem this project sits in.
Base model framework	PyTorch	Required by TorchXRayVision.
Pretrained CXR models	TorchXRayVision — <code>densenet121-res224-nih</code> (fast) and <code>resnet50-res512-all</code> (accurate)	Off-the-shelf, open, no retraining required to get started; see §6 for detail and caveats.
Image processing (Quality/Repair Agents)	OpenCV + scikit-image	Laplacian variance, non-local means denoising, CLAHE, Wiener deconvolution all have direct implementations here.
OOD statistics	NumPy / SciPy (Mahalanobis distance, covariance estimation)	No extra ML framework needed beyond feature extraction via the base model.
Decision Agent v2 (learned variant)	scikit-learn (logistic regression / shallow MLP)	Small, interpretable, fast to iterate on; avoids over-engineering the one genuinely novel component.
Pipeline orchestration	Plain Python pipeline / lightweight task graph (e.g., a simple custom orchestrator, or Prefect/ Airflow if the project scales)	Given only 7 sequential/parallel steps, a hand-rolled orchestrator is likely sufficient for a research prototype; avoid over-tooling.

Experiment tracking	Weights & Biases or MLflow	For threshold calibration runs, ablations, and the per-action accuracy breakdown in §7.
API/serving layer (if a demo is needed)	FastAPI	Lightweight REST endpoint to submit an image and get back a prediction + reliability label + reasoning trace.
Report generation (optional, downstream of Decision Agent)	An open LLM (e.g., GPT-Neo, following the PneumoNet precedent) purely for turning structured output into a readable report — not used for the diagnosis itself.	
Containerization	Docker	For reproducible environment across dev/eval machines.
Dataset handling	pandas (for <code>Data_Entry_2017.csv</code>), Kaggle CLI or NIH box.com download scripts	See §6 for dataset structure.
Visualization / eval dashboards	Matplotlib/Seaborn or a simple Streamlit dashboard	For ROC curves used to calibrate thresholds, and for the action-breakdown accuracy table in §7.

Explicitly not needed: a general-purpose multimodal LLM in the diagnostic loop. Literature evidence (§8) shows GPT-4o correctly identifies only ~69% of pathologies against expert radiology findings — a concrete, cited reason to keep LLM usage confined to optional report-

writing, not decision-making.

6. Dataset

Primary: NIH ChestX-ray14

Detail	Value
Total images	112,120 frontal-view chest X-rays
Unique patients	30,805
Labels	14 categories (text-mined from radiology reports): Pneumonia, Atelectasis, Cardiomegaly, Effusion, Infiltration, Mass, Nodule, Pneumothorax, Consolidation, Edema, Emphysema, Fibrosis, Pleural Thickening, Hernia
Official split	75,312 train / 25,596 test
Source	NIH Clinical Center box.com repository (official); Kaggle "NIH Chest X-rays" mirror (pre-organized)
Structure (Kaggle)	images_001–images_012 PNG folders + Data_Entry_2017.csv (filename, labels, patient ID, age, gender, view position)
Format	PNG, originally 1024×1024, resized to 224×224 or 512×512 for model input

`densenet121-res224-nih` is trained exclusively on this dataset (clean in-distribution reference). `resnet50-res512-all` is trained on NIH + CheXpert + PadChest + MIMIC — which is deliberately useful: CheXpert/PadChest images become a **real, natural OOD test set** for the OOD Agent, rather than requiring synthetic distribution shift.

Secondary datasets from the reference literature (useful for cross-

dataset generalization/ablation): the 5,856-image pneumonia set used in the EfficientNetV2L benchmark study, and the three-dataset validation setup used by PnetoNet.

7. Evaluation Plan

- Accuracy/error rate broken down **by action taken** (Accept/Repair/ Escalate/Reject) — accuracy on the "easy, accepted" subset alone is not sufficient.
 - OOD detection AUROC via the NIH-vs-CheXpert/PadChest natural OOD setup.
 - Calibration error — confidence numbers should be honest, not just top-1-label accuracy.
 - Compute cost per input (fast-path vs. escalated-path).
 - Repair recovery rate, measured empirically: $\text{recovery_}\% = \frac{(\text{variance_after_repair} - \text{variance_blurred})}{(\text{variance_original} - \text{variance_blurred})} \times 100$ on deliberately-corrupted-but-known-correct images. Expect heavier degradation to be less recoverable — report honestly.
 - False-positive OOD rate on **rare-but-valid** cases specifically, separate from true distribution-shift cases (a known OOD-detector failure mode).
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8. Related Work / Positioning

- **MedRAX** (2025): orchestrates multiple existing CXR tools + a multimodal LLM without additional training; introduced ChestAgentBench (2,500 queries, 7 categories); shown to outperform general-purpose and biomedical specialist models on complex CXR queries.
- **RadAgents** (2026): seven agents mirroring the clinical ABCDE review scheme, plus an Orchestrator and a Synthesizer that resolves cross-tool conflicts with traceable logs; an 8B open-

weight VLM structured this way matched or beat GPT-4o and specialist models.

- **ChatGPT-4o radiology accuracy study** (2025): correctly identified 69% of pathologies overall (66% on CXRs specifically) vs. expert radiologists across 500 X-rays — direct evidence for keeping general LLMs out of the diagnostic decision path.
 - **Where this project differs:** MedRAX and RadAgents decompose *diagnostic reasoning itself* (organ-by-organ, tool-by-tool). This project decomposes **trustworthiness reasoning around a single classifier** — quality, novelty, and confidence as independent, explainable signals — treating "should we trust this prediction, and what do we do if not" as the object of study. It is smaller, cheaper, and more explainable by design than either MedRAX or RadAgents.
 - Other supporting literature: EfficientNetV2L/EfficientNetB0 CNN benchmarks (base-model accuracy comparisons, with a caution that near-100% single-dataset numbers can signal overfitting/leakage); PneumoNet's AlexNet+GPT-Neo pattern (precedent for LLM-as-report-writer, not diagnoser); DenseNet201+XAI (SHAP/LIME/Grad-CAM) work as prior art for the Quality/Repair preprocessing choices; and a 2024 survey flagging dataset bias, explainability, class imbalance, and adversarial robustness as open problems this project partially addresses.
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9. Honest Limitations (state proactively)

1. Repair is a proxy for image quality improvement, not proof of diagnostic accuracy — mitigated by restricting repair to reversible defects, tagging repaired outputs, and empirically measuring recovery rate.
2. All thresholds in this document are starting points requiring per-dataset calibration via ROC analysis.
3. OOD detectors can conflate rare-but-real presentations with true distribution shift — report both failure modes separately.
4. No component is clinically validated or regulatory-approved; this is a research prototype, stated plainly wherever demoed.

5. General-purpose LLMs are not reliable enough (69% pathology-ID accuracy in cited literature) to sit in the diagnostic path — hence they are scoped out of this design except as an optional report-writing layer.
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10. Anticipated Review Questions

- **"How do you validate thresholds aren't arbitrary?"** → ROC-curve calibration on held-out validation splits.
 - **"How do you know OOD flags aren't just rare-but-real cases?"** → Acknowledged limitation; false-positive OOD rate reported separately on rare-but-valid cases.
 - **"What if repair/escalation makes things worse?"** → Verification Agent explicitly checks; no improvement past threshold → escalate further, never silently accept.
 - **"What's the system-level evaluation metric, not just base-model accuracy?"** → Action-broken-down accuracy, OOD AUROC, calibration error, and compute cost (§7).
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11. Open Questions / Risks

- Decision Agent v2 (learned) needs a concrete ablation showing it beats the fixed rule table on the accuracy/cost/safety trade-off before it can be claimed as the paper's contribution — not yet demonstrated in source material.
- High reported accuracies in some cited CNN benchmarks (e.g., 99.78%) may reflect dataset leakage rather than true generalization; worth independently verifying rather than citing at face value.
- Repair Agent's reversibility assumption needs explicit bounds — how much blur/noise is "reversible enough" to attempt repair vs. going straight to Reject.